

JINGZHANG RISING

A Century of Self-Reliance, Rising in the Shape of .

A3 Design Booklet · A3

Centennial Jing-Zhang AI Innovation Belt Urban Design Open Call

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Core Judgment: All spatial metrics recomputed in EPSG:4548.

Provisional Boundary: Conceptual recommendations only, not statutory control.

AI Public Governance: Human Fallback Channel + Public Report Card required.

1. Strategic Positioning: China's AI Origin Corridor

In 1909, the Beijing-Zhangjiakou Railway was built as China's first trunk railway designed and constructed independently. Zhan Tianyou created the '?'-shaped switchback at Qinglongqiao, turning the Chinese character for 'people' into a spatial symbol of engineering self-reliance. More than a century later, this exact corridor hosts the highest density of top-tier AI researchers, leading models, and computing resources in China.

Our proposal elevates the historical spirit: from 'independently building railways' to 'independently creating AI intelligence', and from 'human-scale switchback' to 'human-centric upward innovation'.

2. Core Structure: One Spine, Three Terraces, Two Wings, Seven Stitches

- ? One Main Spine: 9.0 km continuous Jing-Zhang Heritage Park greenway + dedicated commuter bicycle expressway.
- ? Three Terraces: Zhongzhiyuan (AI Acceleration), AI Origin Works (Incubation & Open Source), Dazhongsi (Green Compute Cluster).
- ? Two Wings: Zhongguancun Tech Service Wing (west, fundamental research) + Xiaoyuehe Scenario Wing (east, full-domain testing).
- ? Seven Stitches: 7 east-west pedestrian stitch links healing railway severance and reconnecting university campuses.

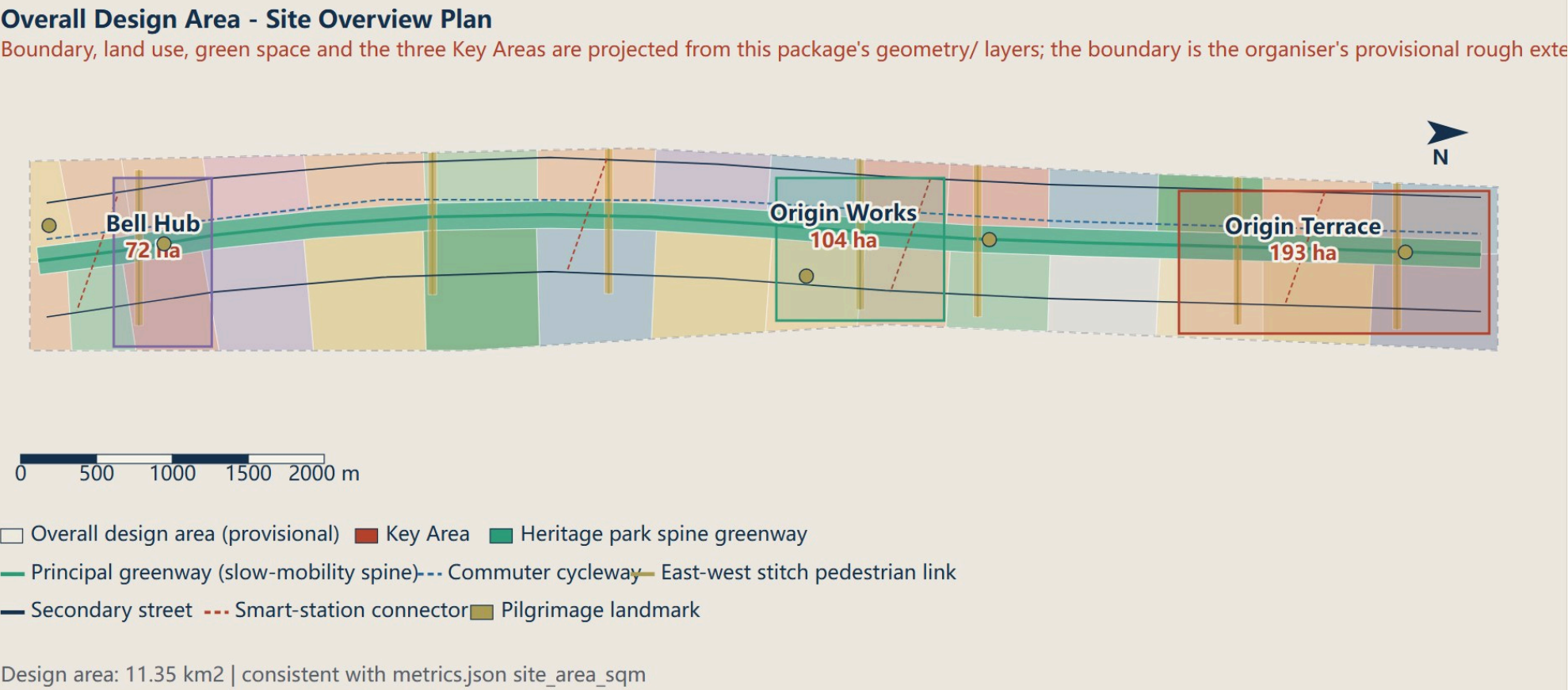


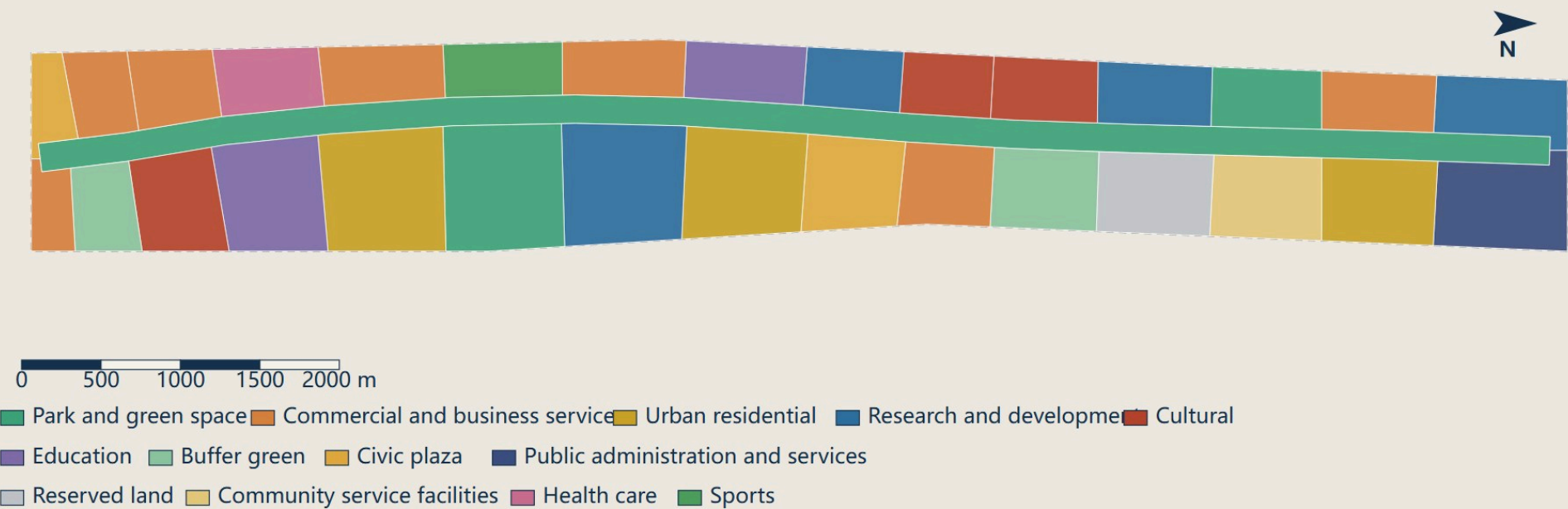
Fig.01 Structure & Landmarks

The overall design scope spans 11.41 km² along the historical railway corridor.

- Key Elements:
- ? 4 Pilgrimage Landmarks:
 - Renzi Terrace (1909, Qinghe)
 - Gaokao Gate (1949, Tsinghuayuan)
 - Clock of Compute (1980, Dazhongsi)
 - Origin Stele (2026, Origin Works)
 - ? Spatial Spine: Jing-Zhang Heritage Park.
 - ? Regional Linkage: Connecting Huairou Science City, Future Science City, and Yizhuang BDA.

Land-Use Structure Plan (block level)

31 parcels coloured by land-use code; legend and area statistics recomputed from land_use.geojson.



Land-use area statistics (recomputed from land_use.geojson)

Code	Land-use category	Area (ha)	Share
1401	Park and green space	257.1	22.5%
05	Commercial and business services	169.7	14.9%
0701	Urban residential	147.0	12.9%
0802	Research and development	138.7	12.2%
0803	Cultural	81.8	7.2%
0804	Education	73.1	6.4%
1402	Buffer green	55.1	4.8%

Code	Land-use category	Area (ha)	Share
1403	Civic plaza	49.4	4.3%
08	Public administration and services	45.7	4.0%
16	Reserved land	36.5	3.2%
0702	Community service facilities	35.9	3.1%
0806	Health care	26.4	2.3%
0805	Sports	25.0	2.2%

Total 1141.3 ha (31 parcels)

Fig.02 Land-Use Partition

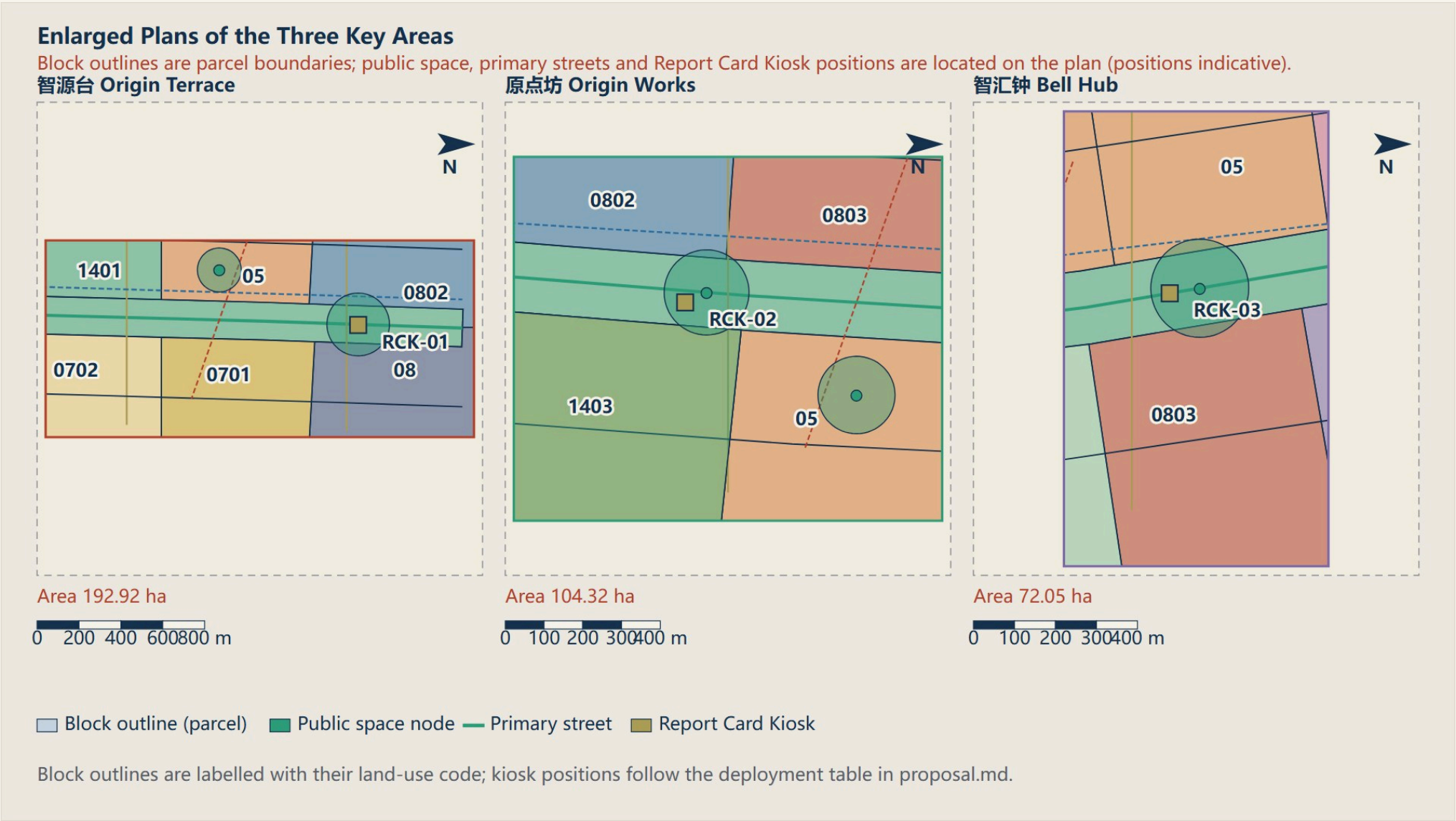
Full spatial partitioning into smart station and innovation anchor catchments.

- Planning Principles:
- ? 4 Smart Station TODs: 500m pedestrian catchment covering major employment centers.
 - ? 14 Functional Anchors: Full coverage without gaps or overlapping polygons.
 - ? Mixed Use: Ground-floor public interfaces, shared maker spaces, and open research courtyards.

Fig.03 Key Areas & Blocks

Detailed block-scale design for the three key areas totaling 369.3 ha.

Three Terraces Depth:
? Zhongzhiyuan (193 ha): ZZ-A..E, FAR 1.6?3.5, Height 24?60 m, Coverage ?35%.
? Origin Works (104 ha): YD-A..E, FAR 1.2?2.8, Height 18?45 m, Coverage ?35%.
? Dazhongsi (72 ha): DZ-A..E, FAR 1.8?3.6, Height 24?60 m, Coverage ?35%.
? Kiosk Deployment: RCK-01, RCK-02, RCK-03 situated at central pedestrian plazas.



Slow Mobility, Blue-Green and Water Overlay Plan

Cycling network, pedestrian priority, green corridors and waterfront are overlaid on one base plan; each layer is listed in the legend below.

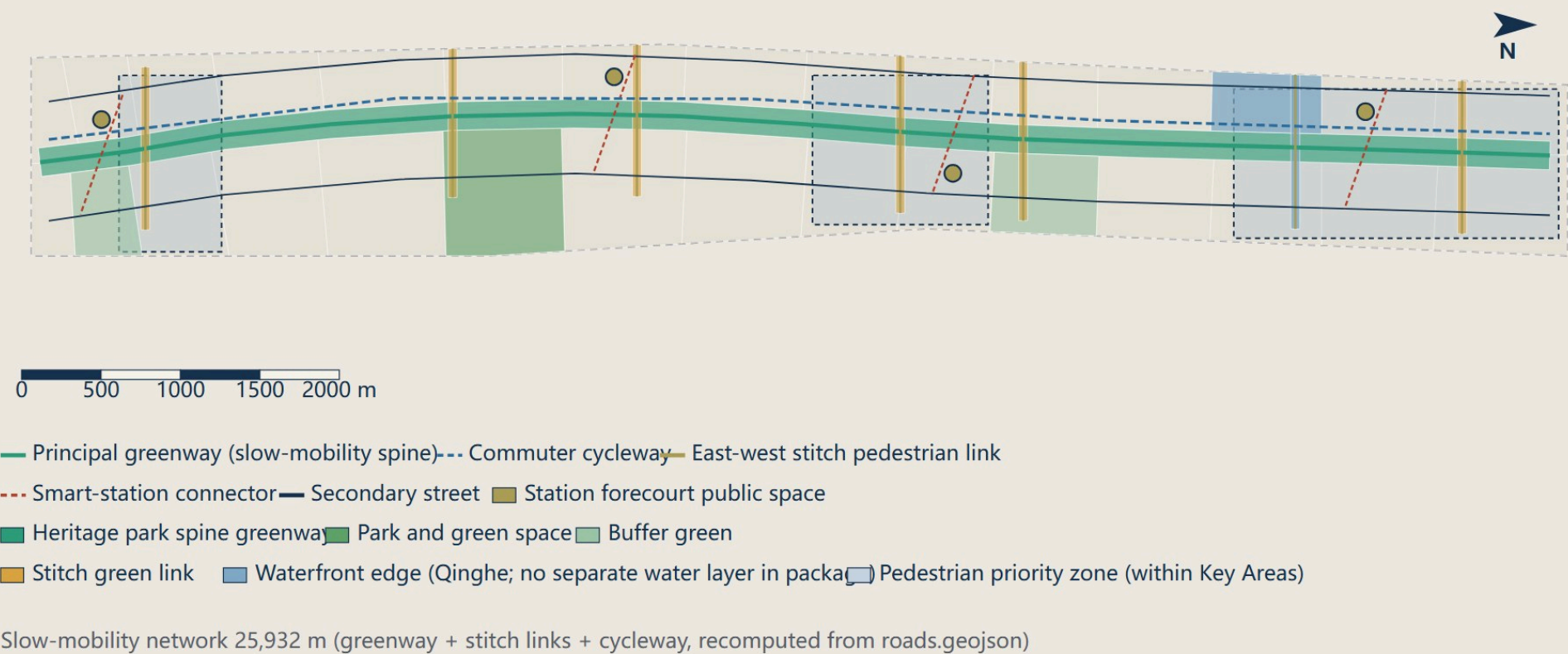


Fig.04 Mobility & Ecology

Comprehensive pedestrian, bicycle, and ecological park corridor network.

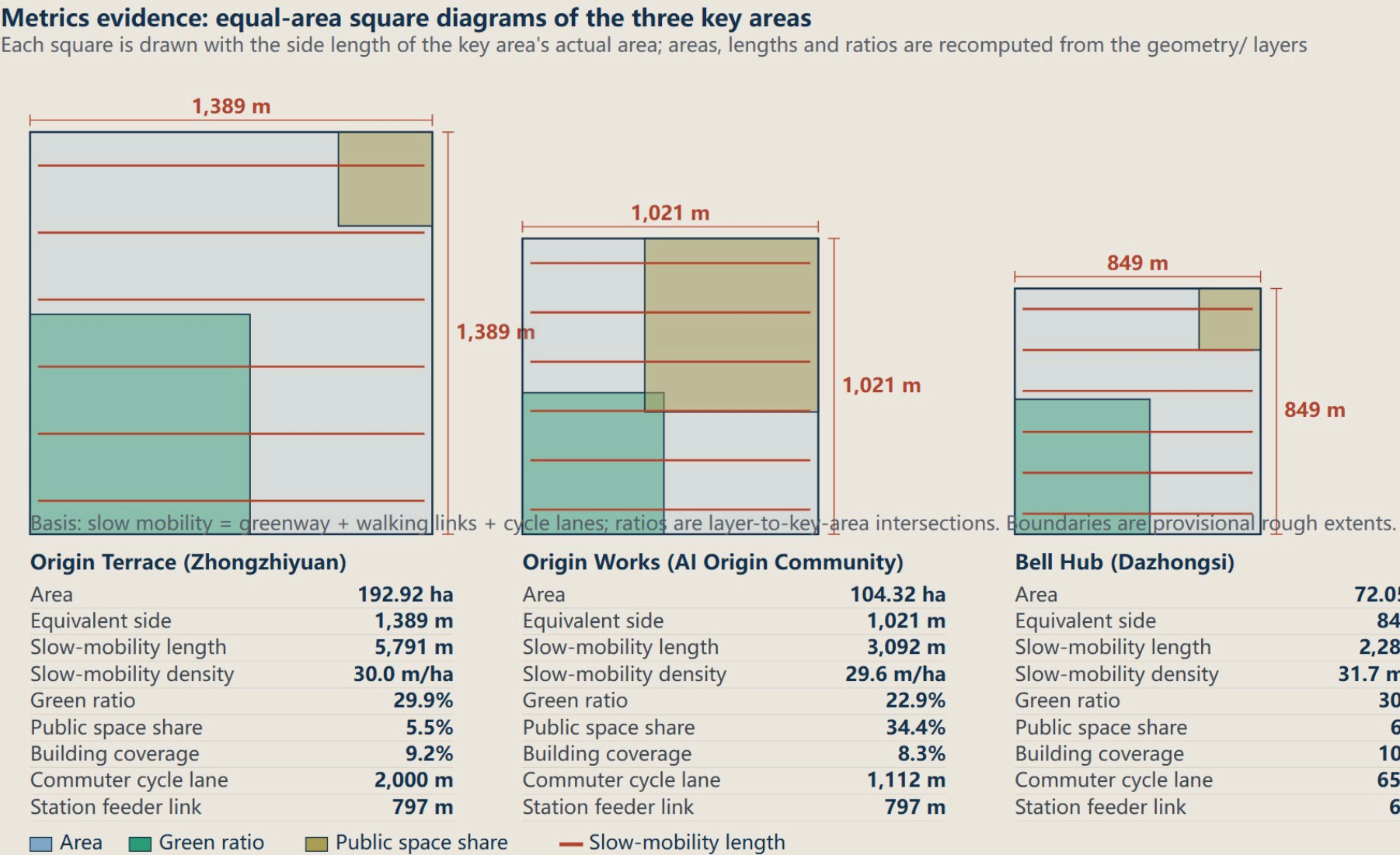
Key Systems:

- ? 9.0 km Main Greenway Spine with 4.5m paved multi-use trail.
- ? 7 East-West Stitch Links over former rail barriers.
- ? Street Sections: Detailed cross-sections for pedestrian (3-5m), bike (2.5-3m), and green verges.
- ? Smart Station Curbsides: Time-of-day dynamic allocation + EV charging ?40%.

Fig.05 Metrics & Governance

Structured evidence chain recomputing 33 core metrics in EPSG:4548.

- Metrics Invariants:
- ? 30 Known Metrics: Statement Value ? Automated Recomputation Value.
 - ? 3 Statutory Controls: FAR, building height, total GFA kept strictly 'unknown'.
 - ? Kiosk Typology: Footprint 3.96 m² (1800x2200x2800 mm), clear zone ?2.0m, wheelchair turning radius ?1.5m.
 - ? Verification: 100% PASS on all deterministic gates.



1. Six Talent Personas

- ? Origination Scientist: Quiet, long-term exploration in dedicated labs with low electromagnetic noise.
- ? Open-Source Engineer: Low-cost flexible co-working, visible contributions recognized on Origin Stele.
- ? AI Startup Founder: Fast iteration, seamless access to test sandboxes and venture capital.
- ? Industry Engineer: Real-world industrial data, edge deployment environments.
- ? International Talent: Bilingual signage, seamless international life services.
- ? Local Community Resident: Primary beneficiary of green parks, smart services with human fallback.

2. 12 AI Scenarios & 4 Sandboxes

- ? 12 Scenario Cards: Covering intelligent traffic dispatch, adaptive park lighting, AI health kiosks, environmental microclimate sensing, and smart street furniture.
- ? 4 Industrial Test Sandboxes:
 - 1. Embodied Robotics Sandbox (Zhongzhiyuan)
 - 2. Autonomous Delivery & Logistics Corridor (Heritage Park)
 - 3. Generative Urban Design Sandbox (Origin Works)
 - 4. Green Compute & Energy Dispatch Sandbox (Dazhongsi)

1. Kiosk Spatial Typology

? Dimensions: 1,800 x 2,200 x 2,800 mm | Footprint: 3.96 m² | Independent detachable base.

? Dual Operating Modes:

- Powered Mode: 32-inch high-brightness outdoor display (1500 nit) with real-time F1?F6 status.
- No-Power Mode: A3 stainless steel etched nameplate + physical QR code pointing to static manifest.

? Universal Accessibility:

- Clear operational zone ?2.0 m, wheelchair turning radius ?1.5 m.
- Screen center height 1,000?1,200 mm for seated/standing accessibility.

2. Six-Item Public Report Card Rules

- ? F1: Algorithm Registration & Government Filing (Critical)
- ? F2: Public Space Scenario Whitelist Permit (Critical)
- ? F3: Dedicated Human Fallback Channel (Critical - AI never sole entry)
- ? F4: Privacy Edge Anonymization Protocol (Critical - No cloud face storage)
- ? F5: Real-Time Energy & Green Compute Traceability (Improvement)
- ? F6: Public Feedback & Direct Appeal Channel (Improvement)

Traffic Light Governance:

- ? Green: Full Operational Access (All F1?F6 verified)
- ? Yellow: 7-day rectification (F5 or F6 pending)
- ? Red: Immediate shutdown & switch to human service (Any F1?F4 missing)

1. Three-Phase Implementation

- ? Phase I (2026?2028) - Origin Initiation:
Greenway spine connection, Open Source Hall, Renzi Terrace, first batch of 4 sandboxes.
- ? Phase II (2029?2031) - Smart Mesh Expansion:
Full 7 stitch links, 4 Smart Station TODs, full coverage of Report Card Kiosks.
- ? Phase III (2032?2035) - Global Leadership:
World-class self-reliant AI ecosystem, international governance dialogue platform.

2. Standards & Compliance Matrix

- ? Ministry of Housing Urban Design Management Measures
- ? Detailed Planning Drafting Guidelines
- ? Universal Accessibility Design Standard (GB 55019-2021)
- ? Interim Measures for Generative AI Services
- ? Elder-Friendly Smart Technology Solutions
- ? Strict Privacy Protocol: No raw biometric face data transmitted to cloud.